

Mini-Design Challenge II: Prototyping an AI Council to Support PBL

Process Documentation Statement:

My council roles are based on the different subject areas, learning objectives and phases of the project. Different members of the council will be available at different phases of the project. The workflow for the council interactions is as follows:

- Phase 1: Cultural Ethnographer. Handoff: Once team completes the Community Chart, they are directed to proceed to Phase 2: Interstellar Linguist and Scientific Historian. Handoff: The linguist analyzes the draft list of artifacts to check for diversity of voices and prompts to consider how to present information to extraterrestrial entities. The Historian helps teams consider some of practical logistics of their choices, including choice of media and format, to ensure information is accessible many thousands of years into the future. > Phase 3: Logistics Architect. The logistics lead helps teams with the mathematical modeling involved in determining their storage limits. It can also encourage mathematics connections to other topics and provide support. > Phase 4: The Launch Visionary. The Visionary helps teams bring their vision together into a final product and can provide design and presentation guidance to the committee.

As teams progress, they retain access to all previous members for consult, as they may need to refine or revise their project or plan. The teams will be given access to the AI Council after working on each phase, as well as 1-2 times during the phases. They will not have continuous access to the Council, but can submit specific questions via the teacher if they feel they need guidance and have hit a road block.

To build my council, I create five separate personae on Google Gems that use Google Gemini. To create a simple UI, I used Cursor to build a web interface that puts each Gem on one screen with a prompt input. Because Gems cannot yet be embedded directly, or have multi-Gem capability on the same screen, the Gem's instructions are pasted into my web app's code, along with a Gemini API that uses Gemini Flash. As a result, the responses from the members are not as consistent and high quality as asking the Gems directly in the Google interface. To help with this, I added more detail into the context of the web app for each persona. This made up for some of the context that the Gems had within the Google interface (on Google platform, the Gems have access to a NotebookLM notebook I have that contains project documents and conversations about project planning).

To improve I would further tune each member persona. I have indirect ways for the members to communicate with each other - in the interface, after a response, a user has the option to send a response to another council member, and keep a running log of responses. I also need to refine responses to be more level-appropriate. Sometimes models give a deluge of information that is difficult for children to organize and process.

Outline of my AI process for this stage:

- Upload project information and background articles and documents in NotebookLM (PBL information, standards documents for ELA, math, science, and social studies, the Dorchester Project Brief as an example, and my Project Brief).
- I asked for outlines of possible AI council members based on my description of the project and examples within the documents. I gave guidance to include information about possible standards to address and asked for explicit and implicit learning that could happen in different subject area.
- I created the 5 council member personas in Google Gems, editing and adding onto the NotebookLM suggestions.
- I asked Cursor for advice on how to integrate Google Gems into a simple web-based UI with a prompt input. Cursor described a method to manually enter persona instructions for each Gem, along with access to the project documents, and a Gemini API.
- I tested the web UI and refined the UX and flow, adding options to save and pass on responses.